

Surejkrishna Melattinkara Sunil

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Education

M.S. Mechanical Engineering (GPA: 3.58)

UC San Diego, June 2026

B.S. Mechanical Engineering

UC San Diego, June 2025

Relevant Coursework: Conduction & Convection Heat Transfer, Fluid Mechanics, Energy Systems, Thermal-Fluids Laboratory, Processing Polymers & Composites

Experience

Undergraduate Researcher, TEMP lab UCSD

December 2024 – June 2025

- Led design, modeling, and testing of a novel wearable thermoelectric liquid cooling module, achieving a 70% increase in cooling across 6 iterations
- Developed and tested a cooling-loop test system by integrating thermocouples and flow sensors, sizing tubing, and selecting the pump through thermal and hydraulic analysis, achieving 16°C maximum temperature drop
- Designed and fabricated FDM 3D-printed flexible heat sinks using Fusion 360, optimizing print parameters to improve in-plane thermal conductivity by 37%
- Performed thermal and fluid calculations in MATLAB to size components, predict performance, and guide design
- Modeled steady-state thermal performance in Ansys to estimate heat dissipation limits and improve future heat sink design

R&D Mechanical Engineering Intern, Bharat Fritz Werner Limited

July 2024 – August 2024

- Led feasibility study to replace air blast oil coolers with phase change material (PCM) systems under 597 W heat loads under continuous operation in high temperature environments
- Estimated system heat loads using hand heat transfer calculations with inputs derived from vendor specifications and on-machine measurements to define cooling requirements for a PCM concept
- Evaluated PCM thermal properties and identified promising non-toxic, non-corrosive candidates aligned with CNC operational needs
- Proposed semi-passive PCM system to senior management, defining constraints and direction for future development

Undergraduate Researcher, Marko Lubarda Lab

May 2023 – June 2024

- Led development of a CAE simulation model for self-locking brakes in Altair Inspire, modeling slider motion and stability under 3 variable loads
- Resolved contact and constraint issues to achieve realistic physics behavior, documenting simulation errors and causes
- Utilized parametric CAD simulations to identify a critical loading condition and failure mode, driving analytical revisions
- Co-authored two journal manuscripts and presented results at the UCSD Lab Expo

Projects

Mechanical Design & Fabrication Lead - Senior Capstone, UCSD Central Research Facility

- Developed a robotic system to automate grabbing, flipping, and transferring mice cages onto a conveyor belt, obtaining 64% of manual throughput for under \$6.5k in 15 weeks
- Designed, sourced, and assembled the Ball Screw linear translation system for automated cage handling
- Created full CAD assemblies in Onshape for manufacturability and integration
- Selected gantry motors and drivers, verifying specs using machine design principles and load calculations
- Machined custom aluminum/steel components on 3-axis mills and waterjet

ANSYS Fluent CFD Ventilation Analysis - Computational Fluid Dynamics

- Built and ran a 3D ANSYS Fluent airflow simulation of a mechanically ventilated classroom to evaluate ventilation effectiveness, low-velocity regions, and aerosol transport behavior
- Ran a 3-level mesh refinement study and post-processed velocity contours and particle pathlines to identify recirculation zones, assess solution sensitivity, and support safer seating-layout recommendations
- Set up steady incompressible flow physics using mass-flow inlet/outlet conditions, a coupled pressure-based solver, and the k- ω SST turbulence model for HVAC-scale indoor airflow analysis

Skills

Manufacturing: 3D printing(FDM), CNC machining(CAM), Waterjet, Manual milling & lathe, laser cutting

CAD: SolidWorks, Fusion 360, Onshape, Autodesk Inventor

Simulation & Analysis: MATLAB, ANSYS, Solidworks, ANSYS Workbench

Mechanical Design: Tolerance analysis, Bill of materials, Design for manufacturing, Thermal system prototyping, DFMA, Finite Element Method

Hobbies

Indoor Rock Climbing, Hiking, Reading